

THE CORE PROBABILITY AXIOMS

**Module 1: The Illusion of Certainty **

1. THE SAMPLE SPACE (Ω)

Before we can calculate the odds of an event, we must map the universe of everything that *could* happen. This boundary is called the Sample Space.

- **Sample Space (Ω):** The complete set of all possible outcomes of a random experiment.
 - *Coin Toss:* $\Omega = \{Heads, Tails\}$
 - *Six-Sided Die:* $\Omega = \{1, 2, 3, 4, 5, 6\}$
- **Event (A):** A specific outcome (or set of outcomes) we care about. It is a subset of the sample space ($A \subseteq \Omega$).
- **Fundamental Probability Formula:** For equally likely outcomes, the probability of Event A occurring is:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes in } \Omega}$$

2. THE THREE AXIOMS OF PROBABILITY

In 1933, Andrey Kolmogorov established the absolute baseline of modern probability. Every statistical calculation in the universe obeys these three unbreakable rules:

Axiom 1: Non-Negativity

The probability of any event is always a real number greater than or equal to zero. You cannot have a negative chance of an event occurring. Absolute zero is the floor.

$$P(A) \geq 0$$

Axiom 2: Unitarity (The Law of Certainty)

The probability that *something* in the sample space will happen is exactly 1 (or 100%).
The universe must choose an outcome.

$$P(\Omega) = 1$$

Axiom 3: Additivity (Mutually Exclusive Events)

If two events cannot possibly happen at the exact same time (they are “mutually exclusive,” meaning their intersection is empty: $A \cap B = \emptyset$), then the probability of either one happening is just the sum of their individual probabilities.

$$P(A \cup B) = P(A) + P(B)$$

3. THE LAW OF LARGE NUMBERS (LLN)

Human intuition fails because we judge probability based on small sample sizes. The math of certainty only reveals itself at scale.

- **The Concept:** As the number of identical, independent trials (N) grows closer to infinity, the experimental, real-world results will converge exactly on the theoretical mathematical probability.
- **The Formula:** If N_A is the number of times Event A occurs in N trials, the limit as trials approach infinity equals the true probability of A :

$$\lim_{N \rightarrow \infty} \frac{N_A}{N} = P(A)$$

4. DETERMINISTIC VS. STOCHASTIC SYSTEMS

Why do we need probability in the first place?

- **Deterministic Systems:** If you know the exact starting conditions, you know the exact final outcome. (e.g., Calculating the trajectory of a bowling ball using Classical Physics).

- **Stochastic (Random) Systems:** The starting conditions are too complex or fundamentally chaotic. We cannot predict a single outcome, but thanks to the Axioms and the Law of Large Numbers, we can perfectly predict the *distribution* of outcomes over time.